

Revision Record						
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Report– R-002

Outboard Power for Homebuilt Boats

I get questions all the time about maximum and minimum sized outboards for homebuilt boats. These things are rather hard to determine unless you know a lot about the boat, the type of water, and the boat owner's preferences. There are certain guidelines however that you can calculate to determine the maximum power based on both the US and Canadian Coast Guard recommendations.

The Coast Guard calculations are based on boat hull shape, boat length and the boat's transom width. It is explained in the US Coast Guard's publication, Safety Standards for Backyard Boat Builders. I provide a link to a pdf file of their publication on the [Free E-Books, Videos and Reports](#) page of the Spira International Web Site. In addition to horsepower recommendations, the Safety Standards also include recommended flotation, loading, identification, ventilation, and other safety recommendations for homebuilt boats. It's a worthwhile read for anyone considering building a boat.

For those who are math challenged, I've run the basic calculations for my power boat designs, in a spreadsheet, and also added my recommended power for these designs. Some boats, such as the Carolinian and Tillamook offer two calculation methods. The first is for flat bottom. This implies a 15" transom opening and tiller steer. The 20T / Remote is the calculation for boats using a standard shaft motor, 20" high opening in the transom, and remote steering, such as a center console or pilothouse.

Boat Max Horsepower Calculation

Based on US and Canadian Coast Guard Recommendations

Spira Int'l Boat Name	Calculation Method	Overall Length	Width Transom	Calculated Factor	CG Rec Max HP	Spira Rec. HP
Alamitos	Remote / 20T	15.0	5.7	85.5	81	40
Albion	Flat Bottom	18.9	6.8	128.5	49	40
Albion	Remote / 20T	18.9	6.8	128.5	167	65
Anacapa	Remote / 20T	19.1	7.0	133.7	177	75
Back Bay	Flat Bottom	7.5	2.4	18.0	N/A	2.5
Bahaman	Flat Bottom	27.0	6.4	172.8	71	65
Bayou Jubilee	Remote / 20T	18.0	5.1	91.8	31	15
Caladesi	Flat Bottom	20.0	5.5	110.0	40	25
Caladesi	Remote / 20T	20.0	5.5	110.0	130	75
Cape Cod	Other	15.2	2.3	35.0	3	2.5
Carolinian	Flat Bottom	18.1	5.0	90.5	30	20
Carolinian	Remote / 20T	18.1	5.0	90.5	91	40
Chesapeake	Remote / 20T	24.0	5.6	134.4	179	90
Clemente	Remote / 20T	24.0	6.4	153.6	217	120
Crawdad	Other	14.2	4.2	59.6	23	9.9
Farallon	Remote / 20T	23.6	7.3	172.3	255	100
Fiesta Bay	Other	13.0	6.0	78.0	37	20
Galveston	Remote / 20T	18.5	6.8	125.8	162	65
Hatteras	Remote / 20T	19.0	5.6	106.4	123	75
Havasus	Other	10.0	4.5	45.0	11	9.9
Hudson	Flat Bottom	15.8	4.9	77.4	24	9.9
Kachemak	Remote / 20T	17.0	5.0	85.0	80	50
Key Largo	Flat Bottom	24.0	5.6	134.4	52	50
Key Largo	Remote / 20T	24.0	5.6	134.4	179	90
Kodiak	Remote / 20T	32.0	10.0	320.0	550	175
Marsh Rat	Other	12.9	3.1	40.0	7	5
Maxi Miss	Flat Bottom	14.1	4.0	56.4	13	9.9
Mini Miss	Flat Bottom	7.0	2.4	16.8	N/A	2.5
Minutemen	Flat Bottom	13.3	3.5	46.6	8	5
Mission Bay	Flat Bottom	11.0	4.0	44.0	7	5
Oscoda	Remote / 20T	16.1	6.7	107.9	39	25
Oysterman	Flat Bottom	16.1	4.2	67.6	19	15
Pescadero	Flat Bottom	14.2	4.2	59.6	15	9.9
Sabine	Remote / 20T	19.0	6.3	119.7	45	30
Seneca	Flat Bottom	14.0	5.0	70.0	20	15
Seneca	Remote / 20T	14.0	5.0	70.0	50	40
Sitka	Remote / 20T	27.0	8.0	216.0	342	120
Tillamook	Flat Bottom	16.5	5.7	94.1	32	20
Tillamook	Remote / 20T	16.5	5.7	94.1	98	50
Winchester	Remote / 20T	23.2	7.0	162.4	235	75
Wye River	Flat Bottom	16.2	5.7	92.3	31	20
Wye River	Remote / 20T	16.2	5.7	92.3	95	50